

The Fragile Epistemic Foundations of Representative Democracy

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Abstract

We develop a model in which an incumbent politician can exert costly effort to learn about an uncertain and policy-relevant state of the world before proposing a policy to a representative voter. The politician does not perfectly observe the voter's prior belief (or opinion). His epistemic effort increases the probability of re-election through two channels: a *quality* effect, as better information improves the match between the policy and the state; and a *hegemony* effect, as it induces the voter to place less weight on her prior and thereby compresses ex post disagreement across voter types. We show that, because of the hegemony effect, when the politician has more precise information about the voter, he implements a less informed policy. We connect this result to the recently observed rise in the use of big data in politics.

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1 Introduction

Modern societies face complex problems, such as climate change, AI regulation, and increased geopolitical tensions, often in the face of uncertainty. A long tradition in the social sciences, dating back to [Condorcet \[1785\]](#), argues that democracy has an epistemic comparative advantage over other political systems, and is therefore better equipped to tackle such problems ([Piketty \[1999\]](#); [List and Goodin \[2001\]](#)). The intuition is that when each citizen holds a piece of information and all share the same objective, voting *horizontally* aggregates the fragmented pieces and society converges to the truth.

However, a *vertical* dimension also exists. Citizens may delegate information acquisition to their representatives due to asymmetric costs of learning. The informational quality of representative democracy then crucially depends on elected politicians' incentives to acquire policy-relevant information. We study this problem in a context where politicians are office-motivated but hold imperfect knowledge of voters' prior belief.

We consider a setting with one (representative or pivotal) voter (she) and one (incumbent) politician (he). Both players are uncertain about a state of the world (e.g., the pace of global warming, the risks of a new AI, or the intentions of a rival nation). Facing an election, the politician needs to propose a policy to the voter. The voter would like the policy to match the state, and chooses whether to re-elect the politician based on the proposed policy or to replace him with a challenger. The politician wants to be re-elected.

We depart from standard models of electoral accountability under policy uncertainty (e.g., [Canes-Wrone et al. \[2001\]](#); [Maskin and Tirole \[2004\]](#)) in two important ways. First, we allow for the possibility that, before the election game, the politician exerts costly effort to learn about what constitutes good policy. Formally, he observes a signal about the state of the world. The precision of the signal is endogenous and depends on the politician's effort. More precise signals are more costly. At the same time, by exerting more effort, the politician can improve his (and the voter's) estimate of the state of the world and thus propose a better policy.

Second, we assume that the politician does not perfectly observe (the mean of) the voter’s prior belief about the state of the world.¹ We call this mean the voter’s *opinion*, as it essentially measures what each possible voter type believes to be the best policy absent any new information transmitted by the politician. In this sense, we call *moderate* a voter whose opinion is close to the mean of the distribution of opinions in the politician’s mind, and *extremist* a voter whose opinion is far from it. After observing any signal revealed by the politician, the voter updates her beliefs about the state of the world. As the voter is Bayesian, her posterior estimate of the state (and thus her preferred policy) is a function of the newly transmitted information and of her prior opinion. It follows that estimating the voter’s prior belief has a value for the politician because it allows him to propose a more targeted policy.

We start our equilibrium analysis by characterizing the politician’s optimal choice of a policy, holding the quantity of information fixed (Proposition 1). The voter wants the policy to match the state, and the politician tries to please the voter in an attempt to be re-elected. The politician’s optimal policy is thus a linear combination of the average prior opinion across all possible voter types and of the acquired signal, with weights that depend on the precision of the signal relative to that of the prior.

When the voter is more *stubborn* (i.e., when she is more confident about her opinion, or, formally, when her prior belief is highly concentrated around the mean), her posterior estimate of the state stays close to the prior, and so does the implemented policy. By contrast, when the voter is *open-minded*, the implemented policy relies more on the acquired information.

If our reasoning were to stop here — that is, if we disregard the choice of learning effort — the politician’s uncertainty over the voter’s opinion entails only costs for the voter. Because the politician targets the average type, a voter with a very extreme opinion, who believes the state to be very high or very low, pays a higher cost. Consequently, an extremist voter is *ex ante* more likely to punish the incumbent and elect the challenger.

¹For recent empirical evidence that politicians do not perfectly know their constituents’ views see e.g. [Broockman and Skovron \[2018\]](#) and [Otjes and Rasmussen \[2026\]](#).

We then characterize the incumbent politician’s equilibrium epistemic effort (Proposition 2). We show that the marginal electoral benefit of information acquisition can be decomposed into two interpretable channels.

The first is a *quality* effect. The politician exerts effort as a more precise signal improves the match between the proposed policy and the true state. All voter types benefit equally from this.

The second, more interesting channel is a *hegemony* effect. By acquiring and providing information to the electorate, the politician induces the voter to place less weight on her prior belief and compresses ex post disagreement across voter types. All voter types’ preferred policies move toward the signal transmitted by the politician, who, in this sense, exercises a hegemony over the electorate. Since the politician must please a voter whose opinion is unknown, the hegemony is valuable: less disagreement across types translates into a smaller expected distance between the proposed policy and the policy preferred by the voter’s *true* type.

Importantly, we find that this second channel is stronger when the politician is more uncertain about the voter’s prior belief. In such cases, disagreement across possible voter types is larger, and reducing such disagreement through a more informative signal yields a greater electoral benefit. Therefore, when the politician can observe the voter’s opinion with higher precision, he implements less informed policies. If his knowledge of the voter is precise enough, the equilibrium features no learning and the policy is only a function of the average opinion.

We connect this result to the growing use of voter data in politics. Digital platforms and data brokers allow campaigns to build individual-level datasets from public records, consumer data, and online activity. Campaigns use these data to *profile* voters and estimate their opinions with unprecedented precision (Hersh [2015]; Nickerson and Rogers [2014]; Aridor et al. [2024]). The conventional wisdom is that better knowledge of voters improves representation (Butler and Nickerson [2011]). Our paper shows that this argument may be too optimistic, as it overlooks politicians’ incentives to acquire information in the first place.

After proving our main proposition, we discuss some additional results.

First, we study how the voter’s stubbornness affects the politician’s learning effort. Increased stubbornness has two opposite effects. On the one hand, it lowers the weight the voter places on the signal, thus making learning less valuable for the politician. On the other hand, since each voter type’s preferred policy stays closer to her own opinion, stubbornness also increases disagreement across voter types, thereby strengthening the hegemony effect. We show that when the politician’s uncertainty about the voter’s opinion is low enough, the first effect dominates, so that increased stubbornness reduces learning. Moreover, a sufficiently stubborn voter always induces the politician to stop learning (the equilibrium epistemic effort is zero).

Second, we study welfare. While better profiling always reduces the epistemic quality of representation, the voter may be partially compensated due to improved targeting. We characterize the solution to this trade-off by comparing *perfect* and *imperfect* profiling. A politician who perfectly observes the voter’s prior opinion proposes, for any given signal, exactly the voter’s preferred policy. While this is good for the voter, perfectly knowing the voter’s type eliminates the hegemony effect, thereby leading to a less informed policy choice. We show that which effect dominates depends on the voter’s type. Without profiling, the politician caters to the average opinion. Therefore, a moderate voter (whose opinion is close to the average) loses from profiling, while a voter with an extreme opinion gains. By contrast, the politician always benefits from better profiling.

Third, we show that our results are robust to the possibility that the politician could strategically decide to hide the acquired information. Hiding the signal makes the voter more uncertain about the state and leads her to rely more on the prior, which increases ex post disagreement across voter types. Hiding thus weakens both the quality and the hegemony effect, and the politician never finds it optimal.

Finally, we discuss some applications of our model beyond politics. We highlight how our results apply to any setting in which an agent can invest effort to learn about a state of the world and is then evaluated by a principal whose prior is her private information.

The remainder of the paper is organized as follows. In the next section, we discuss our contribution to the literature. Section 3 illustrates the model setup. We present our main results in Section 4. Section 5 discusses comparative statics, welfare, strategic hiding of information, and applications beyond politics. Section 6 concludes. Proofs are in Appendix A.

2 Related Literature and Contribution

A long tradition defends democracy on epistemic grounds. Since Condorcet [1785], the argument is that voting aggregates dispersed private signals (Feddersen and Pesendorfer [1997]; List and Goodin [2001]; Estlund [2008]). A related literature endogenizes information acquisition (Persico [2004]; Martinelli [2006]; Gerardi and Yariv [2008]; Koriyama and Szentes [2009]). In this tradition, knowledge comes from voters. By contrast, in our model, information is acquired by politicians and transmitted to citizens.

We model information acquisition in democracy as a vertical delegation problem. Our model thus belongs to the political agency tradition (Barro [1973]; Ferejohn [1986]; Canes-Wrone et al. [2001]; Maskin and Tirole [2004]; see Ashworth [2012] for a review). This literature sees voters and politicians as principals and agents, respectively. Closest to us are models in which an office-motivated incumbent chooses between the policy he believes to be right and the policy voters expect. This literature takes politicians' information as given and asks whether they have an incentive to *transmit* it (Heidhues and Lagerlöf [2003]; Millner et al. [2020]; Kartik et al. [2026]). We focus instead on politicians' incentives to *acquire* information in the first place.

Finally, we relate to the literature on social media and big data in politics. This literature studies how platforms shape voters' preferences, often through misinformation and polarization (Allcott and Gentzkow [2017]; Levy [2021]; Germano et al. [2026]; see Aridor et al. [2024] for a review),² and how campaigns build data on voters (Hersh [2015]) and use it to target messages at the individual level (Fowler et al. [2021]; Titova [2026]).

²Social media may also have positive effects on democracy, for instance by facilitating coordination among voters in autocratic contexts (Enikolopov et al. [2020]).

We point to a different and overlooked channel. We argue that the digital revolution may not only have affected how voters *form* their political preferences, but also what politicians *know* about these preferences. We find that this additional knowledge may undermine the fragile epistemic foundations of representative democracy.

3 Model

There are two players, one representative voter (she) and one incumbent politician (he). There is a state of the world, $\omega \in \mathbb{R}$, which is unknown to both players. The voter would like the proposed policy, $x \in \mathbb{R}$, to match the state. The incumbent politician commits to a policy before the election. The voter, who observes the proposed policy, chooses whether to re-elect the politician ($v = I$) or to elect a challenger ($v = C$). For simplicity, we assume that the voter's payoff from electing the challenger is captured by a fixed outside option $k < 0$, which is the voter's private information. Therefore, the voter's utility function is:

$$u(v, x) = \begin{cases} -(x - \omega)^2 & \text{if } v = I \\ k & \text{if } v = C \end{cases}.$$

From the incumbent politician's perspective, the variable k is uniformly distributed, i.e., $k \sim \mathcal{U}[-\Delta k, 0]$, where $\Delta k > 0$.

The voter's prior belief over the state, ω , is normally distributed: $\omega \sim \mathcal{N}(\mu, 1/\rho)$. We refer to the prior mean, μ , as the voter's *opinion*. The voter's opinion is her private information. In particular, the politician believes that μ may take one of finitely many values, μ_i , with $i = 1, \dots, N$ (and $N \geq 2$), each with probability $\pi_i > 0$, and $\sum_i \pi_i = 1$.³ We often refer to μ_i (or, simply, to i) as the *type* of the voter. The parameter $\rho > 0$ measures the voter's *stubbornness*, i.e., the confidence in her own opinion. We assume ρ is the same for every type of voter, and known to the politician.

Before committing to a policy, x , the politician decides how much effort to exert to

³We assume that the *true* voter's opinion satisfies $\mu \in \{\mu_1, \dots, \mu_N\}$.

learn about the state of the world. Formally, he observes a signal:

$$s = \omega + \epsilon , \tag{1}$$

with $\epsilon \sim \mathcal{N}(0, 1/\rho_\epsilon)$. The parameter $\rho_\epsilon \geq 0$ captures the politician's *epistemic effort*, i.e., the effort exerted to learn about the state. If $\rho_\epsilon = 0$, the signal is pure noise and carries no information. Effort, however, is costly: acquiring information of precision ρ_ϵ entails a cost $C(\rho_\epsilon) = c\rho_\epsilon$, where $c > 0$. The politician is office-seeking. He obtains a payoff of one if re-elected and zero otherwise, net of the cost of effort. His utility function is therefore:

$$u_P(v, \rho_\epsilon) = \begin{cases} 1 - C(\rho_\epsilon) & \text{if } v = I \\ -C(\rho_\epsilon) & \text{if } v = C \end{cases} .$$

For a given level of effort, the politician thus maximizes his probability of re-election.

Let S denote the set of all possible signals. Once received, both the signal, $s \in S$, and its precision, ρ_ϵ , are publicly revealed.⁴

We summarize the timing of the game and the information structure as follows.

Stage 1 Nature independently determines the voter's type, i , and the state of the world, ω .

The incumbent politician chooses his level of effort, ρ_ϵ , observed by the voter.

Stage 2 The politician and the voter observe the realization of the signal, s . The politician chooses a policy, x .

Stage 3 The voter observes x and decides whether to re-elect the incumbent. Payoffs are realized.

Our solution concept is Perfect Bayesian Equilibrium in pure strategies. We require strategies to be optimal given beliefs, and beliefs to be updated according to Bayes' Rule.

⁴We assume that the signal is *hard* information, i.e., information which, once obtained, is hard to hide or manipulate. While this assumption is helpful to convey our results, it is not consequential (see Section 5.3).

4 Equilibrium

4.1 The Politician's Optimal Policy

In this section, we characterize the incumbent politician's optimal choice of a policy, x^* . We proceed by backward induction, starting from [Stage 3](#).

A voter of type i re-elects the incumbent if and only if $\mathbb{E}[u(I, x) \mid s, i] \geq k$. The incumbent's probability of re-election when facing a voter of type i is:

$$\Pr(k \leq \mathbb{E}[u(I, x) \mid s, i]) = \max \left\{ 0, \frac{\mathbb{E}[-(x - \omega)^2 \mid s, i] + \Delta k}{\Delta k} \right\}, \quad (2)$$

where $\mathbb{E}[-(x - \omega)^2 \mid s, i]$ is the expected utility of a voter of type i , given her posterior estimate of the state.

In [Stage 2](#), the incumbent chooses x to maximize his expected probability of re-election. Since (2) is non-decreasing in the voter's expected utility, and⁵

$$\mathbb{E}[-(x - \omega)^2 \mid s, i] = -(x - x_i^*(s))^2 - \text{Var}(\omega \mid s, i), \quad (3)$$

where $x_i^*(s) \equiv \mathbb{E}[\omega \mid s, i]$, the politician's problem reduces to

$$\min_{x \in \mathbb{R}} \sum_{i=1}^N \pi_i (x - x_i^*(s))^2. \quad (4)$$

Proposition 1 derives the politician's optimal policy as a function of the signal.

Proposition 1. *Let $\alpha \equiv \frac{\rho_\epsilon}{\rho_\epsilon + \rho}$ and $\bar{\mu} \equiv \sum_{i=1}^N \pi_i \mu_i$. For a given signal, s , the politician's optimal policy is:*

$$x^*(s) = \alpha s + (1 - \alpha) \bar{\mu}. \quad (5)$$

The interpretation is simple. By standard properties of normally distributed information, the posterior estimate of the state of a voter of type i is a convex combination of the observed signal and her opinion, where the weights depend on the relative precisions

⁵See the proof of Proposition 1 in the Appendix.

of the signal and the prior:

$$x_i^*(s) \equiv \mathbb{E}[\omega \mid s, i] = \alpha s + (1 - \alpha) \mu_i . \quad (6)$$

This is the policy preferred by a voter of type i . When the voter is confident in her prior belief (high ρ), her preferred policy remains close to the prior, μ_i . If, on the other hand, the voter sees that the politician has exerted a lot of effort (high ρ_ϵ), her preferred policy relies more on the acquired information (i.e., on the signal, s).

The politician does not know the voter's type. Therefore, his optimal policy, $x^*(s) = \sum_i \pi_i x_i^*(s)$, is the average of the preferred policies across all possible voter types and it coincides with the preferred policy of a voter whose opinion is the *average opinion*, $\bar{\mu}$.

At this stage of the game, the politician's uncertainty generates only costs for the voter. We illustrate this by comparing the voter's equilibrium payoff from re-electing the incumbent with the benchmark case where the politician observes μ_i (a case we refer to as *perfect profiling*). Under perfect profiling, the politician would choose the policy $x_i^*(s)$, and the voter's utility from electing the incumbent would be equal to:

$$\mathbb{E}[-(x_i^*(s) - \omega)^2 \mid s, i] = -\text{Var}[\omega \mid s, i] = -\frac{1}{\rho + \rho_\epsilon} . \quad (7)$$

By contrast, when the politician does not observe μ_i (*imperfect profiling*) and chooses $x^*(s)$, using (3), we can write voter i 's expected utility as:

$$\mathbb{E}[-(x^*(s) - \omega)^2 \mid s, i] = -[x^*(s) - x_i^*(s)]^2 - \frac{1}{\rho + \rho_\epsilon} , \quad (8)$$

which reduces to:

$$-\left(\frac{\rho}{\rho + \rho_\epsilon}\right)^2 (\mu_i - \bar{\mu})^2 - \frac{1}{\rho + \rho_\epsilon} . \quad (9)$$

Under imperfect profiling, the voter suffers a utility loss that depends on the distance between her prior opinion, μ_i , and the average opinion, $\bar{\mu}$. Notice that, because the politician maximizes the payoff of the average voter's type, a more extremist voter is

more penalized. Consequently, an extremist voter is ex ante (before k is drawn) more likely to punish the incumbent and support the challenger.

This cost is weighted by how much the voter is attached to her opinion, which is measured by $1 - \alpha = \rho/(\rho + \rho_\epsilon)$. When the voter is willing to change her mind, priors do not matter much, and the cost of extremism is lower. In other words, when the voter's stubbornness is lower, the posterior estimates of the state are closer across all possible voter types. This also makes the job easier for the incumbent politician. As we will see in the following subsection, this plays an important role in determining the politician's optimal choice of effort.

4.2 The Politician's Optimal Epistemic Effort

In this section, we characterize the incumbent politician's optimal choice of effort, ρ_ϵ^* , in [Stage 1](#). Using Equation (9), we can re-write the incumbent's expected probability of re-election given $x^*(s)$ as:

$$\frac{-\left(\frac{\rho}{\rho + \rho_\epsilon}\right)^2 \sum_i \pi_i (\mu_i - \bar{\mu})^2 - \frac{1}{\rho + \rho_\epsilon} + \Delta k}{\Delta k} . \quad (10)$$

We restrict our attention to cases such that the voter's outside option is low enough that (10) lies between zero and one.⁶ The incumbent politician chooses $\rho_\epsilon \geq 0$ to maximize his expected utility evaluated at $x^*(s)$:

$$\mathbb{E}[u_P(v, \rho_\epsilon)] = \frac{1}{\Delta k} \left[-\left(\frac{\rho}{\rho + \rho_\epsilon}\right)^2 \sum_i \pi_i (\mu_i - \bar{\mu})^2 - \frac{1}{\rho + \rho_\epsilon} + \Delta k \right] - c\rho_\epsilon . \quad (11)$$

Our main result is:

Proposition 2. *Let $\sigma_\mu^2 \equiv \sum_{i=1}^N \pi_i (\mu_i - \bar{\mu})^2$ and $\underline{\sigma}^2 \equiv \frac{c\Delta k \rho^2 - 1}{2\rho}$. The politician's optimal effort, ρ_ϵ^* , is unique. It is zero if $\sigma_\mu^2 \leq \underline{\sigma}^2$, and it is positive and strictly increasing in σ_μ^2 if $\sigma_\mu^2 > \underline{\sigma}^2$.*

⁶Formally, we assume $\Delta k > 1/\rho + \max_{i,j} (\mu_i - \mu_j)^2$. The numerator of (10) is increasing in ρ_ϵ , so it is enough to check it at $\rho_\epsilon = 0$, where it equals $\Delta k - \sum_i \pi_i (\mu_i - \bar{\mu})^2 - 1/\rho$. Since $(\mu_i - \bar{\mu})^2 \leq \max_j (\mu_i - \mu_j)^2$ for every i , the assumption implies that the numerator is always positive. It is smaller than Δk because the first two terms are negative.

The parameter σ_μ^2 measures how uncertain the politician is about the voter's opinion.

Figure 1 illustrates how it is constructed.

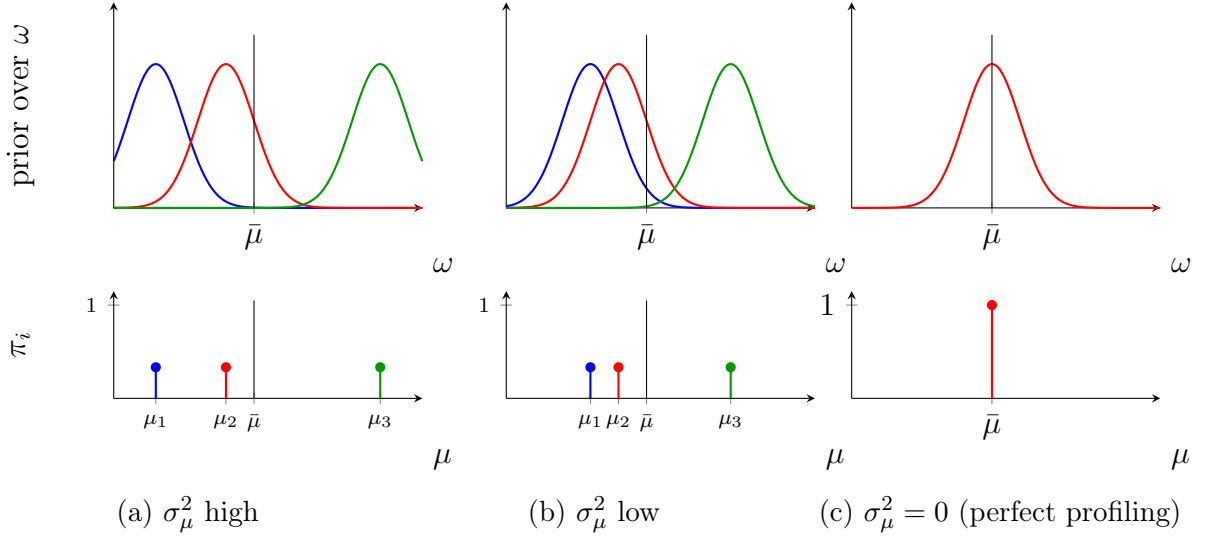


Figure 1: **The politician's uncertainty over the voter's opinion.**

Note: Top: the voter's possible priors over the state. Bottom: the distribution of the voter's opinion from the politician's perspective. The average opinion, $\bar{\mu}$, is the same in all panels.

Proposition 2 shows that the more uncertain the politician is about the voter's opinion, the higher the level of effort he will exert, and the more adequate the policy will be. Moreover, when his knowledge of the voter's opinion is precise enough, he stops learning altogether.

We view this result to be of particular interest given the recent advent of big data and profiling technologies in politics. The increased profiling power of platforms (which we model in reduced-form as a reduction in σ_μ^2), by enabling politicians to observe voters' data and opinions, decreases representatives' incentives to exert effort to learn which policies are optimal, thereby lowering the epistemic quality of representation.

The intuition behind this result is as follows. An office-motivated politician exerts effort because information about the state increases his probability of re-election. Let $B(\rho_\epsilon)$ denote the politician's benefit from effort, i.e., the term in square brackets in (11).

Its derivative, the *marginal benefit* of effort, is the sum of two terms:

$$B'(\rho_\epsilon) = \underbrace{\frac{1}{(\rho + \rho_\epsilon)^2}}_{\text{quality}} + \underbrace{\frac{2\rho^2\sigma_\mu^2}{(\rho + \rho_\epsilon)^3}}_{\text{hegemony}}. \quad (12)$$

Each term corresponds to one of the two components of the voter's expected utility in (9). We now discuss each in turn.

- (i) The first term is the derivative of $-1/(\rho + \rho_\epsilon)$, the expected loss due to the politician's residual uncertainty over the state. We call it the *quality effect*. More precise information improves the match between the policy and the true state. All voter types like this.
- (ii) The second term is the derivative of $-(1 - \alpha)^2\sigma_\mu^2$, the expected loss due to the politician's uncertainty about the voter's opinion. We call it the *hegemony effect*. When the politician is better informed about the state, the voter puts less weight on her own opinion and more on the signal. As a result, the preferred policies of all voter types move toward the signal (and the implemented policy).

Figure 2 illustrates the hegemony effect. It shows the voter's possible posterior means over the state after the same signal, s , for a low and a high level of effort exerted by the politician.

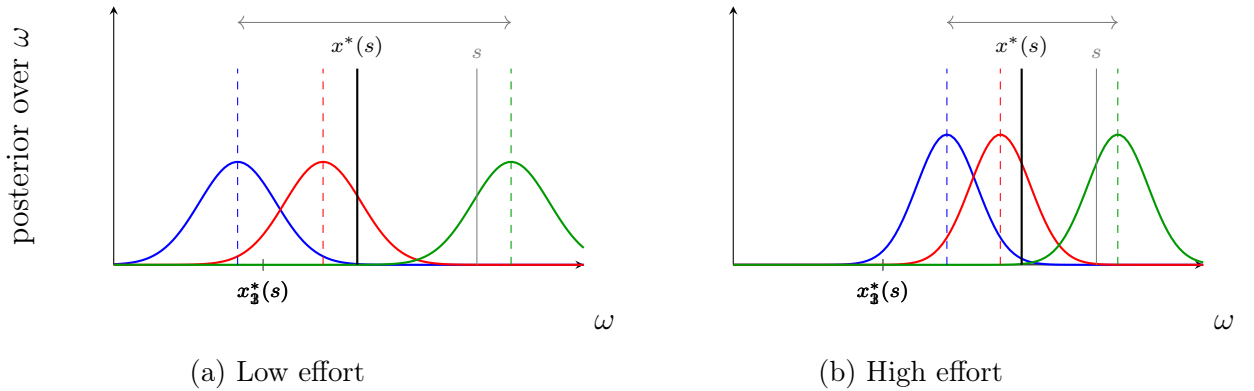


Figure 2: **The hegemony effect.**

Note: Posteriors over the state of the three types of Figure 1(a), after the same signal, s , for a low (a) and a high (b) level of effort. Dashed lines: the preferred policies of the three types. Solid line: the politician's policy.

While the quality effect does not depend on σ_μ^2 , and it is therefore the only reason

to exert effort for a politician who perfectly knows the voter’s opinion, the hegemony effect is stronger when the politician is more uncertain about the voter’s opinion. This happens because a higher dispersion in the voter’s possible preferred policies increases the expected marginal benefit for the politician of reducing such dispersion by communicating a more precise signal on the state of the world.

Therefore, better profiling weakens the hegemony effect and lowers the politician’s incentives to exert effort. The optimal level of effort is zero when, evaluated at $\rho_\epsilon = 0$, the two effects together do not cover the marginal cost of learning.

5 Discussion

In this section, we offer a brief discussion of several aspects of our main result, Proposition 2. We study how the voter’s stubbornness affects learning (Section 5.1), the welfare consequences of profiling for the voter and for the politician (Section 5.2), the robustness of our result to strategic hiding of information (Section 5.3), and applications beyond politics (Section 5.4).

5.1 Comparative Statics: Stubborn Voters

The comparative statics of the politician’s optimal level of effort, ρ_ϵ^* , with respect to the voter’s stubbornness, ρ , is ambiguous. As ρ increases, the importance of the signal for the voter’s posterior estimate of the state, α , decreases (see Equation (6)). This lowers the politician’s incentives to exert effort to obtain a better signal.

However, a countervailing effect may emerge. For a given revealed signal, a higher level of stubbornness also increases the *disagreement* about the optimal policy among all possible voter types. As we have seen, this strengthens the hegemony effect, thereby increasing the politician’s incentives to exert effort (see the discussion following Proposition 2).

The countervailing effect works through the hegemony effect. It is therefore weak when the politician is not sufficiently uncertain about the voter’s opinion. The following

corollary shows that, in this case, the first effect always dominates.

Corollary 1.

- (i) Fix σ_μ^2 . There exists $\bar{\rho} > 0$ such that the politician does not learn ($\rho_\epsilon^* = 0$) if and only if $\rho \geq \bar{\rho}$.
- (ii) Fix $\rho < \sqrt{\frac{2}{c\Delta k}}$. There exists $\bar{\sigma}_\mu^2$ such that, if $\underline{\sigma}_\mu^2 < \sigma_\mu^2 < \bar{\sigma}_\mu^2$, the optimal effort, $\rho_\epsilon^* > 0$, is decreasing in ρ .

Corollary 1 contains two results. The first statement confirms that a sufficiently stubborn voter always induces the politician to stop learning. When ρ is large enough, the voter's preferred policy hardly depends on the signal (α is close to zero), the hegemony effect vanishes, and the chosen policy collapses to the average opinion, creating a world in which the politician caters to voter priors rather than to the actual state of the world.

The second statement shows that, when the politician's uncertainty over the voter's opinion is sufficiently low, the hegemony effect is too weak to counteract the first effect, and facing a more stubborn voter always induces the politician to exert less effort.

5.2 Welfare: Misaligned Incentives

Our main result, Proposition 2, shows that when politicians have more precise information over voters' opinions, they implement less informed policies. However, it is silent about welfare effects. This is relevant because — as already discussed in Section 4.1 — better profiling can also improve the voter's welfare. We now show that this is not always true once politicians' learning incentives are taken into account.

We measure the welfare of a voter of type i by her equilibrium expected utility from re-electing the incumbent, $\mathbb{E}[u(I, x^*(s)) \mid s, i]$, i.e., by Equation (9) evaluated at $\rho_\epsilon = \rho_\epsilon^*$. By (9), this quantity does not depend on the realized signal.⁷ Better profiling affects the voter's welfare in two ways. First, by Proposition 2, it reduces the politician's learning effort, ρ_ϵ^* . This hurts every type of voter. Second, it may change the distance between

⁷The voter can also elect the challenger. Her overall expected payoff is thus $\mathbb{E}_k[\max\{\mathbb{E}[u(I, x^*(s)) \mid s, i], k\}]$, which is increasing in $\mathbb{E}[u(I, x^*(s)) \mid s, i]$, so all comparisons below are unaffected.

the voter's opinion and the average opinion, $|\mu_i - \bar{\mu}|$, to which the politician caters. This benefits a voter of type i if it brings $\bar{\mu}$ closer to μ_i , and hurts her otherwise.

To study the interplay between the two effects, we compare two regimes. We say that profiling is *perfect* when the politician knows the voter's opinion, so that $\mu_i = \bar{\mu}$ and $\sigma_\mu^2 = 0$; and *imperfect* otherwise. Let $\rho_\epsilon^*(\sigma_\mu^2)$ denote the politician's optimal effort as a function of his uncertainty. Under perfect profiling, the politician chooses the voter's preferred policy, but exerts the lowest possible level of effort, $\rho_\epsilon^*(0)$. Under imperfect profiling, he caters to the average opinion, but, if $\sigma_\mu^2 > \underline{\sigma}_\mu^2$, he exerts positive effort, $\rho_\epsilon^*(\sigma_\mu^2) > \rho_\epsilon^*(0)$.

The following result shows that the voter's preference between the two regimes depends on the distance between the voter's opinion and the average opinion.

Corollary 2. *Let*

$$K(\sigma_\mu^2) \equiv \frac{(\rho + \rho_\epsilon^*(\sigma_\mu^2)) (\rho_\epsilon^*(\sigma_\mu^2) - \rho_\epsilon^*(0))}{\rho^2 (\rho + \rho_\epsilon^*(0))}. \quad (13)$$

(i) *A voter of type i is better-off under imperfect profiling than under perfect profiling if and only if*

$$(\mu_i - \bar{\mu})^2 < K(\sigma_\mu^2),$$

(ii) *If $\sigma_\mu^2 \leq \underline{\sigma}^2$, $\rho_\epsilon^*(\sigma_\mu^2) = \rho_\epsilon^*(0) = 0$, so that $K(\sigma_\mu^2) = 0$ and all types of voter are better-off under perfect profiling. If $\sigma_\mu^2 > \underline{\sigma}^2$, $\rho_\epsilon^*(\sigma_\mu^2) > \rho_\epsilon^*(0)$, so that $K(\sigma_\mu^2) > 0$ and strictly increasing in σ_μ^2 .*

The intuition behind the first statement is simple. Recall that, under imperfect profiling, the politician always implements the policy corresponding to the average opinion. Therefore, a voter whose opinion is close to the average opinion has little to gain from profiling. By contrast, a voter with an extreme opinion is poorly represented under imperfect profiling, and thus gains from perfect profiling, even though the implemented policy is less informed.

The second statement describes how the threshold varies with the politician's uncertainty about the voter's opinion, keeping $\bar{\mu}$ fixed (a mean-preserving spread). When

$\sigma_\mu^2 \leq \underline{\sigma}^2$, the politician's learning effort is zero under both regimes. Imperfect profiling then has only costs, and every voter weakly prefers perfect profiling. By contrast, when $\sigma_\mu^2 > \underline{\sigma}^2$, imperfect profiling improves the epistemic quality of representation. When σ_μ^2 increases, the politician exerts more effort under imperfect profiling. Therefore, the policy quality gap between the two regimes widens, and the set of voter types who are better off under imperfect profiling expands. Since $K(\sigma_\mu^2)$ grows monotonically with σ_μ^2 , a voter with any given opinion prefers imperfect profiling when the politician's uncertainty is large enough.

By contrast, the politician is always better off when his information is more precise.⁸ Formally:

Corollary 3. *The politician's equilibrium expected utility, $\mathbb{E}[u_P(v, \rho_\epsilon^*)]$, is strictly decreasing in σ_μ^2 .*

5.3 Strategic Manipulation of Information

In the baseline version of our model, the signal, s , is publicly observed. We now show that if the politician is given the opportunity to hide the acquired information and to choose the same policy for different signals, he would never find it optimal to do so.

To illustrate this, suppose that there exists a set of signals, $\bar{S} \subseteq S$, such that the politician chooses the same policy, $\bar{x}$, for all $s \in \bar{S}$. After observing $\bar{x}$, the voter infers only that $s \in \bar{S}$, and evaluates the policy accordingly.

The following proposition compares this strategy with full revelation.

Proposition 3. *For any $\bar{S} \subseteq S$ and any $\bar{x}$, the politician's probability of re-election from choosing $\bar{x}$ after every $s \in \bar{S}$ is weakly lower than his probability of re-election from choosing $x^*(s)$ after every $s \in \bar{S}$.*

The intuition mirrors the argument behind our main result, Proposition 1.

First, a voter who only learns that $s \in \bar{S}$ is more uncertain about the state than a voter who observes the revealed signal, s . This weakens the quality effect. Any policy

⁸This is important in light of our connection to the literature on big data in politics. Politicians typically pay digital platforms to obtain voters' data. Coherently, in our model, the politician would have a positive willingness to pay to reduce σ_μ^2 .

$\bar{x}$ chosen after hiding is therefore a worse match for the state, from the voter’s point of view.

Second, when the signal is hidden, the voter uses her prior opinion to *guess* it. In this case, the voter’s preferred policy — which the politician tries to estimate in the policy stage — is then a function of her expectation of the signal (which depends on the prior). Consequently, when the signal is hidden, the voter’s preferred policy depends more on the prior, so that disagreement across voter types is always at least as large as under full revelation. This weakens the hegemony effect.

Combining these two observations, it follows that concealing information cannot be optimal for the politician.

5.4 General Applicability

The logic captured by our model applies more broadly to situations in which an agent can invest effort to learn about a state of the world and is subject to a promotion or retention decision by a principal whose prior over that state is not perfectly known.⁹

A possible application is corporate strategy. Consider a firm that must choose a pricing plan. The CEO delegates this decision to an internal team. The optimal action depends on an uncertain state of the world (e.g., the elasticity of demand), which is unknown to both parties. The principal holds a prior belief about the state, and this belief is private information. Before making a decision, the pricing team can exert costly effort to acquire information, for instance through market research or consultancy. The team privately bears the cost of effort, but would like to pick the best price possible (from the CEO’s point of view).

Another application is academic research. Researchers, like politicians in our framework, must decide how much effort to devote to discovering the true state of the world through rigorous and challenging inquiry, while being evaluated for tenure by committees whose beliefs about what constitutes good research are only imperfectly known.

⁹ Che and Kartik [2009] study a related problem in which a decision maker (DM) may hire an adviser, who can learn about a utility relevant state of the world. They show that it is optimal for the DM to hire an adviser with a prior different from her own. In our model, it is the uncertainty on the principal’s prior which affects the incentives of the agent.

Our results show that voters do not necessarily benefit when politicians know what they think; that CEOs may want to hide their views from their employees; and that academic committees read better research proposals when researchers do not observe their prior beliefs.

6 Conclusions

Since [Condorcet \[1785\]](#), the epistemic argument for democracy has relied on horizontal aggregation of dispersed information. When each citizen receives a private signal about a state of the world, voting combines these signals and ensures society’s convergence to the truth. This is the so-called *wisdom of the crowd*. However, in representative democracies, citizens delegate the acquisition of policy-relevant information to elected officials. The epistemic quality of democracy then depends on a vertical channel: on politicians’ incentives to *acquire* and transmit information to the electorate.

We study this problem in a context where politicians have imperfect information about the electorate’s preferences. We find that reductions in this uncertainty (such as those unlocked by big data and profiling technologies) reduce politicians’ incentives to learn which policies are socially desirable, thereby undermining the epistemic quality of democratic representation.

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A Proofs

Proof of Proposition 1

Consider a voter of type i . For a given policy x and state ω , the voter's quadratic loss $(x - \omega)^2$ can be written as:

$$\begin{aligned} (x - \omega)^2 &= [(x - \mathbb{E}[\omega \mid s, i]) + (\mathbb{E}[\omega \mid s, i] - \omega)]^2 \\ &= (x - \mathbb{E}[\omega \mid s, i])^2 + 2(x - \mathbb{E}[\omega \mid s, i])(\mathbb{E}[\omega \mid s, i] - \omega) + (\mathbb{E}[\omega \mid s, i] - \omega)^2 . \end{aligned}$$

The expected loss is then:

$$\begin{aligned} \mathbb{E}[(x - \omega)^2 \mid s, i] &= \\ &= (x - \mathbb{E}[\omega \mid s, i])^2 \\ &+ 2(x - \mathbb{E}[\omega \mid s, i])\mathbb{E}[\mathbb{E}[\omega \mid s, i] - \omega \mid s, i] \\ &+ \mathbb{E}[(\mathbb{E}[\omega \mid s, i] - \omega)^2 \mid s, i] . \end{aligned}$$

where the second term is equal to zero because $\mathbb{E}[(\mathbb{E}[\omega \mid s, i] - \omega) \mid s, i] = 0$ and the third is equal to $\text{Var}(\omega \mid s, i) = \frac{1}{\rho + \rho_\epsilon}$.

Therefore, voter i 's expected utility from re-electing the incumbent can be written as:

$$\mathbb{E}[-(x - \omega)^2 \mid s, i] = -(x - \mathbb{E}[\omega \mid s, i])^2 - \frac{1}{\rho + \rho_\epsilon} .$$

The politician's problem is:

$$\min_x \left\{ \sum_{i=1}^N \pi_i \left((x - \mathbb{E}[\omega \mid s, i])^2 + \frac{1}{\rho + \rho_\epsilon} \right) \right\} .$$

The FOC reads:

$$\begin{aligned} \frac{\partial}{\partial x} \left\{ \sum_{i=1}^N \pi_i \left((x - \mathbb{E}[\omega \mid s, i])^2 + \frac{1}{\rho + \rho_\epsilon} \right) \right\} &= \\ 2 \sum_{i=1}^N \pi_i (x - \mathbb{E}[\omega \mid s, i]) &= 0 \\ x \sum_{i=1}^N \pi_i - \sum_{i=1}^N \pi_i \mathbb{E}[\omega \mid s, i] &= 0 , \end{aligned}$$

that is:

$$x^*(s) = \sum_{i=1}^N \pi_i \mathbb{E}[\omega \mid s, i] . \quad (14)$$

We are left to compute $\mathbb{E}[\omega \mid s, i]$. By standard properties of normally distributed information,

$$\mathbb{E}[\omega \mid s, i] = \frac{\rho}{\rho + \rho_\epsilon} \mu_i + \frac{\rho_\epsilon}{\rho + \rho_\epsilon} s = \alpha s + (1 - \alpha) \mu_i .$$

Substituting into (14) gives (5). □

Proof of Proposition 2

The derivative of the objective function, (11), with respect to ρ_ϵ is:

$$\frac{B'(\rho_\epsilon)}{\Delta k} - c ,$$

where B' is given by (12). The second derivative is $\frac{B''(\rho_\epsilon)}{\Delta k}$, with

$$B''(\rho_\epsilon) = -\frac{2(\rho + \rho_\epsilon + 3\rho^2\sigma_\mu^2)}{(\rho + \rho_\epsilon)^4} < 0 ,$$

so that the objective function is strictly concave.

If

$$B'(0) = \frac{1 + 2\rho\sigma_\mu^2}{\rho^2} \leq c\Delta k , \quad (15)$$

the derivative is always negative and the unique solution to the politician's problem is a corner, $\rho_\epsilon^* = 0$. This is the case if and only if

$$\sigma_\mu^2 \leq \frac{c\Delta k\rho^2 - 1}{2\rho} ,$$

i.e., if $\sigma_\mu^2 \leq \underline{\sigma}^2$.

Otherwise, there is an interior solution, $\rho_\epsilon^* > 0$. This is implicitly defined by:

$$B'(\rho_\epsilon^*) - c\Delta k = \frac{\rho + \rho_\epsilon^* + 2\rho^2\sigma_\mu^2}{(\rho + \rho_\epsilon^*)^3} - c\Delta k = 0 . \quad (16)$$

Since $B''(\rho_\epsilon^*) < 0$ and

$$\partial B' / \partial \sigma_\mu^2 = \frac{2\rho^2}{(\rho + \rho_\epsilon^*)^3} > 0 ,$$

by the implicit function theorem $\rho_\epsilon^*(\sigma_\mu^2)$ is an increasing function. □

Proof of Corollary 1

We prove each statement of the corollary in turn.

- By the proof of Proposition 2, $\rho_\epsilon^* = 0$ if and only if (15) holds. The LHS is continuous and strictly decreasing in ρ . Then, let $\bar{\rho}$ be the unique value at which it equals $c\Delta k$. This completes the proof.
- Differentiating the first order condition, (16):

$$\frac{\partial B'}{\partial \rho} = \frac{2 [\rho_\epsilon^* (2\rho\sigma_\mu^2 - 1) - \rho (1 + \rho\sigma_\mu^2)]}{(\rho + \rho_\epsilon^*)^4}.$$

If $\sigma_\mu^2 \leq \frac{1}{2\rho} \equiv \bar{\sigma}_\mu^2$, the first term in square brackets is non-positive and the second is negative, so the numerator is negatives. Hence, recalling that $B''(\rho_\epsilon^*) < 0$, by the implicit function theorem $\rho_\epsilon^*(\rho)$ is a decreasing function. The condition $\rho < \sqrt{\frac{2}{c\Delta k}}$ guarantees that $\underline{\sigma}_\mu^2 < \bar{\sigma}_\mu^2$, so that $\rho_\epsilon^* > 0$. $\square$

Proof of Corollary 2

We prove each statement in turn.

- Under perfect profiling, $\sigma_\mu^2 = 0$, $\mu_i = \bar{\mu}$, and

$$\mathbb{E}[u(I, x^*(s)) \mid s, i] = -\frac{1}{\rho + \rho_\epsilon^*(0)}.$$

Under imperfect profiling, $\mathbb{E}[u(I, x^*(s)) \mid s, i]$ is given by (9) evaluated at $\rho_\epsilon = \rho_\epsilon^*(\sigma_\mu^2)$.

The voter is better off under imperfect profiling if and only if

$$\left(\frac{\rho}{\rho + \rho_\epsilon^*(\sigma_\mu^2)}\right)^2 (\mu_i - \bar{\mu})^2 < \frac{1}{\rho + \rho_\epsilon^*(0)} - \frac{1}{\rho + \rho_\epsilon^*(\sigma_\mu^2)} = \frac{\rho_\epsilon^*(\sigma_\mu^2) - \rho_\epsilon^*(0)}{(\rho + \rho_\epsilon^*(0))(\rho + \rho_\epsilon^*(\sigma_\mu^2))}.$$

Multiplying both sides by

$$\frac{(\rho + \rho_\epsilon^*(\sigma_\mu^2))^2}{\rho^2}$$

gives

$$(\mu_i - \bar{\mu})^2 < K(\sigma_\mu^2),$$

with $K(\sigma_\mu^2)$ as in (13). Since $\rho_\epsilon^*(\sigma_\mu^2) \geq \rho_\epsilon^*(0)$ by Proposition 2, $K(\sigma_\mu^2) \geq 0$.

- If $\sigma_\mu^2 \leq \underline{\sigma}^2$, then $\rho_\epsilon^*(\sigma_\mu^2) = \rho_\epsilon^*(0) = 0$ by Proposition 2, so that $K(\sigma_\mu^2) = 0$.

If $\sigma_\mu^2 > \underline{\sigma}^2$, $\rho_\epsilon^*(\sigma_\mu^2)$ is the unique solution of (16). Note that $K(\sigma_\mu^2)$ is increasing in $\rho_\epsilon^*(\sigma_\mu^2)$, which in turns is increasing in σ_μ^2 by Proposition 2. $\square$

Proof of Corollary 3

If $\sigma_\mu^2 \leq \underline{\sigma}^2$, then $\rho_\epsilon^* = 0$. The politician's expected utility, (11), simplifies to:

$$\mathbb{E}[u_P(v, 0)] = \frac{\Delta k - \sigma_\mu^2 - 1/\rho}{\Delta k},$$

which is strictly decreasing in σ_μ^2 .

Otherwise, ρ_ϵ^* is interior and, applying the Envelope Theorem to (11),

$$\frac{\partial \mathbb{E}[u_P(v, \rho_\epsilon^*)]}{\partial \sigma_\mu^2} = \frac{\partial \mathbb{E}[u_P(v, \rho_\epsilon)]}{\partial \sigma_\mu^2} \Big|_{\rho_\epsilon = \rho_\epsilon^*} = -\frac{1}{\Delta k} \left(\frac{\rho}{\rho + \rho_\epsilon^*} \right)^2 < 0. \quad \square$$

Proof of Proposition 3

For a voter of type i , let $m_i \equiv \mathbb{E}[\omega \mid s \in \bar{S}, i]$ and $v_i \equiv \text{Var}(\omega \mid s \in \bar{S}, i)$. By the same decomposition as in the proof of Proposition 1, the voter's expected utility from re-electing the incumbent, given $\bar{x}$, is:

$$-(\bar{x} - m_i)^2 - v_i.$$

We want to show that the average (across types) expected loss of the voter after observing $\bar{x}$ is bounded below by the average expected loss from $x^*(s)$ in (10), i.e.,

$$\sum_i \pi_i [(\bar{x} - m_i)^2 + v_i] \geq \left(\frac{\rho}{\rho + \rho_\epsilon} \right)^2 \sigma_\mu^2 + \frac{1}{\rho + \rho_\epsilon}. \quad (17)$$

Two observations deliver this result. First, using the law of total variance:

$$\begin{aligned} v_i &= \mathbb{E}[\text{Var}(\omega \mid s, i) \mid s \in \bar{S}, i] + \text{Var}(\mathbb{E}[\omega \mid s, i] \mid s \in \bar{S}, i) \\ &= \frac{1}{\rho + \rho_\epsilon} + \text{Var}(\mathbb{E}[\omega \mid s, i] \mid s \in \bar{S}, i) \geq \frac{1}{\rho + \rho_\epsilon}. \end{aligned} \quad (18)$$

Pooling can only increase the voter's uncertainty.

Second, by (6),

$$m_i = \alpha \mathbb{E}[s \mid s \in \bar{S}, i] + (1 - \alpha)\mu_i.$$

Let $g(\mu) \equiv \mathbb{E}[s \mid s \in \bar{S}]$, so that $\mathbb{E}[s \mid s \in \bar{S}, i] = g(\mu_i)$. Note that the function g is non-decreasing: a voter with a higher prior mean expects, conditional on $s \in \bar{S}$, a higher signal. Therefore, for any two types with $\mu_i \geq \mu_j$, we have that:

$$m_i - m_j = \alpha [g(\mu_i) - g(\mu_j)] + (1 - \alpha)(\mu_i - \mu_j) \geq (1 - \alpha)(\mu_i - \mu_j),$$

i.e.,

$$|m_i - m_j| \geq (1 - \alpha)|\mu_i - \mu_j| \quad \forall i, j. \quad (19)$$

For any given pair of types, (i, j) , hiding the signal cannot reduce ex post disagreement between them below its level under full revelation.

We now combine the two observations. Let $\bar{m} \equiv \sum_i \pi_i m_i$ denote the average (across types) posterior estimate of the state. The average, $\bar{m}$, minimizes the expected squared error, so:

$$\sum_i \pi_i (\bar{x} - m_i)^2 \geq \sum_i \pi_i (\bar{m} - m_i)^2.$$

Moreover, expanding the square and after some algebra,

$$\sum_i \pi_i (\bar{m} - m_i)^2 = \frac{1}{2} \sum_i \sum_j \pi_i \pi_j (m_i - m_j)^2.$$

Using (19):

$$\sum_i \pi_i (\bar{x} - m_i)^2 \geq \frac{1}{2} \sum_i \sum_j \pi_i \pi_j (m_i - m_j)^2 \geq \frac{(1 - \alpha)^2}{2} \sum_i \sum_j \pi_i \pi_j (\mu_i - \mu_j)^2 = (1 - \alpha)^2 \sigma_\mu^2.$$

Therefore, by (18):

$$\sum_i \pi_i [(\bar{x} - m_i)^2 + v_i] \geq (1 - \alpha)^2 \sigma_\mu^2 + \frac{1}{\rho + \rho_\epsilon} = \left(\frac{\rho}{\rho + \rho_\epsilon} \right)^2 \sigma_\mu^2 + \frac{1}{\rho + \rho_\epsilon},$$

which is (17). □